



TRAINING PROGRAMME

Linux & Bash Scripting

Work at the Linux command line and automate everyday tasks with Bash

39 HOURS

FOR Students, developers and IT staff starting with Linux



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Overview

This programme takes participants from their first terminal session to Bash scripts that run unattended and fail safely. They learn how a Linux system is organised, how to manage files, users, permissions, processes, services and packages, and how to process text with the standard command-line tools. The second half turns these commands into scripts: variables, conditions, loops, functions, arrays and traps, then error handling, debugging and ShellCheck. Every chapter ends with a lab on a Linux virtual machine, and the programme closes with a final project in which each participant builds and presents a small automation toolkit.

At a glance

#	PHASE	FOCUS	HOURS
01	Linux Essentials	The shell, the filesystem, permissions and text processing	9 h
02	System Administration Basics	Processes, services, scheduling, packages and networking	6 h
03	Bash Scripting	Bash syntax, control flow, functions and robust scripts	12 h
04	Final Project	An automation toolkit, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		39 h

Key facts

AUDIENCE	Students, developers and IT staff starting with Linux
PREREQUISITES	None. Basic computer use; no prior Linux or programming experience needed
DURATION	39 h: 33 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs at the command line
LAB ENVIRONMENT	Runs locally: one laptop per participant (8 GB RAM minimum) with two Linux virtual machines prepared by the trainer, one Ubuntu Server and one Rocky Linux, so both apt and dnf are practised; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can use and manage a Linux system from the command line and write, debug and schedule robust Bash scripts that automate backups, user provisioning and health reports

PHASE 01

9 H

Linux Essentials

Participants learn how a Linux system is organised and how to work efficiently at the command line: finding help, moving through the filesystem, controlling who can read and change files, and chaining small tools into pipelines. Validated lab: a shared project directory with the right owners, groups and permissions, and a one-line pipeline that summarises a log file.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Linux & the Shell	Distributions, kernel vs userspace, terminal and shells, man pages and help	Lecture + Lab	1.5 h
1.2	Filesystem & Navigation	FHS layout, absolute and relative paths, find and locate, file types, hard and symbolic links	Lecture + Lab	1.5 h
1.3	Permissions & Users	rx, chmod and chown, umask, sudo, useradd and groups, /etc/passwd and /etc/shadow	Lecture + Lab	3 h
1.4	Text Processing	cat, less, head and tail, grep, sed, awk, cut, sort, uniq and wc, pipes and redirection	Lecture + Lab	3 h

PHASE 02

6 H

System Administration Basics

A running system is a set of processes and services. Participants inspect and control them, read the logs, schedule jobs, install software and check network connectivity, then connect to another machine with SSH keys. Validated lab: a service installed, enabled and diagnosed from its logs, a cron job, and passwordless SSH access to a second virtual machine.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Processes & Services	ps and top, kill and signals, systemctl, journalctl, cron	Lecture + Lab	3 h
2.2	Packages & Networking Basics	apt and dnf, ip, ss, ping, curl and wget, dig, /etc/hosts, SSH keys and client configuration	Lecture + Lab	3 h

PHASE 03

12 H

Bash Scripting

The commands from the first two phases become scripts. Participants write Bash with correct quoting and exit codes, build control flow, functions and argument parsing, then make their scripts safe to run unattended: strict mode, traps, logging, debugging and ShellCheck. Validated lab: a script that takes options, validates its input, logs what it does and passes ShellCheck without warnings.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Bash Fundamentals	Variables, quoting, exit codes, test and [[]], if and case	Lecture + Lab	3 h
3.2	Loops & Functions	for, while and until, command substitution, arithmetic, functions, arguments and getopts	Lecture + Lab	3 h
3.3	Advanced Bash	Arrays, here-documents, file descriptors, subshells, aliases, trap	Lecture + Lab	3 h
3.4	Robust Scripts	set -euo pipefail, debugging with bash -x, ShellCheck, logging, best practices	Lecture + Lab	3 h

PHASE 04

6 H

Final Project

Each participant builds a small automation toolkit for a provided server, reusing the scripts from the phase labs, then presents it as they would to a system administration team.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Brief & Build	Read the brief, plan the scripts, then write the backup with rotation, the user provisioning from a CSV file and the system health report	Project	3 h
4.2	Schedule, Test & Document	Schedule the scripts with cron, test failure cases, pass ShellCheck, write the README	Project	1.5 h
4.3	Demo & Review	Live demo on a fresh virtual machine; questions and feedback	Project	1.5 h

FINAL PROJECT

Server Automation Toolkit

A set of Bash scripts that back up directories with rotation, create user accounts from a CSV file and produce a daily health report, scheduled with cron, safe to rerun and documented for the next administrator.

STACK Bash, coreutils, grep, sed, awk, tar, cron, systemd journal, ShellCheck, Git

DELIVERABLE Script repository with README, cron schedule, test log, live demo